



Follow the Steps

Quick facilitation notes & answer key

Goal

Students execute a 6-step algorithm in order to fill a row of boxes, producing an AB (circle-square) pattern, then read the result and extend the pattern. Carrying out a sequence of steps to create a mathematical representation (a pattern) is C3.1; predicting the next term is reasoning about the outcome (C3.2).

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil (red and blue crayons optional, to colour ● and ■). Optional: counters to lay out the pattern before drawing.

Run it (about 20 min)

1. I DO: read step 1 and draw ● in box 1. 'I do exactly what the step says, then go to the next step.'
2. WE DO: Exercise 1 — follow steps 2-6 together, drawing ■, ●, ■, ●, ■. Exercise 2 — box 6 is a square ■.
3. YOU DO: Exercise 3 — name the pattern (circle-square / AB) and predict box 7 (●).
4. Connect: 'A computer runs its steps in order, exactly — and the same steps always make the same pattern.'

Answer key (modelled by the run-gate)

Ex 1 — boxes 1-6: ● ■ ● ■ ● ■ (circle, square, circle, square, circle, square).

Ex 2 — box 6 is a square (■).

Ex 3 — the pattern is circle-square, an AB pattern; box 7 would be a circle (●) because the pattern repeats ● then ■, and box 6 ended on ■.

Why it works: each step is carried out in order and only changes its own box, so the row always comes out the same.

Watch for & differentiate

Common slip: skipping a box or doing steps out of order — have students touch each step and its box as they go.

Simplify: pre-draw boxes 1-2 together, then let students finish 3-6.

Extend: ask students to write a step 7 and step 8 that keep the pattern going (● then ■).

Success indicator: the child fills the boxes ● ■ ● ■ ● ■ and names the AB pattern, predicting box 7 as ●.